

3.1.3 Physical training

Students should develop knowledge and understanding of the principles of training and different training methods in order to plan, carry out, monitor and evaluate personal exercise and training programmes.

3.1.3.1 The relationship between health and fitness and the role that exercise plays in both

Content	Additional information
Health and fitness	Definitions of health and fitness.
The relationship between health and fitness	Decreased fitness because of ill health, ie poor health can result in an inability to train, lowers fitness. Increased fitness despite ill health, ie unhealthy but able to train, increases fitness.

3.1.3.2 The components of fitness, benefits for sport and how fitness is measured and improved

Content	Additional information
The components of fitness	Definitions of the following components of fitness: • agility • balance • cardiovascular endurance (aerobic power) • coordination • flexibility • muscular endurance • power/explosive strength (anaerobic power) • reaction time • strength (maximal, static, dynamic and explosive) • speed.
Linking sports and physical activity to the required components of fitness	Understand and justify why the components of fitness (as stated above) may or may not be needed when performing certain physical activities and sports.

Content	Additional information
Reasons for and limitations of fitness testing	Reasons for fitness testing: • to identify strengths and/or weaknesses in a performance/the success of a training programme • to monitor improvement • to show a starting level of fitness • to inform training requirements • to compare against norms of the group/ national averages • to motivate/set goals • to provide variety in a training programme. Limitations of fitness testing: • tests are often not sport specific/too general • they do not replicate movements of activity • they do not replicate competitive conditions required in sports • many do not use direct measuring/sub-maximal – therefore inaccurate/some need motivation/some have questionable reliability • they must be carried out with the correct procedures to increase validity.

Content	Additional information
Measuring the components of fitness	Knowledge of the main procedures of the tests used to measure the following components of fitness: • agility – Illinois Agility Test • balance – Stork Stand Test • cardiovascular endurance (aerobic power) – Multi Stage Fitness Test • coordination – Wall Toss Test • flexibility – Sit and Reach Test • muscular endurance – Sit-Up Bleep Test • power/explosive strength (anaerobic power) – Vertical Jump Test • reaction time – Ruler Drop Test • maximal strength – One Rep Max Test • speed – 30 Metre Sprint Test • strength – Handgrip Dynamometer Test. Testing procedures refers to ‘how each test is carried out’ and includes reference to how the test is organised (when applicable) in relation to the following: • the facilities and the equipment needed to set it up • the procedures that have to be followed – the tasks and the rules • the measurements that are used to score the performance • the way conclusions are drawn from the scores/results. Evaluate whether or not these tests are relevant to performers in different sporting activities.
Demonstration of how data is collected for fitness testing	Understanding of how test scores are measured/recorded (eg in seconds, levels, centimeters, numbers). Definitions of the terms qualitative and quantitative, in relation to the collection of fitness testing data. Understanding that the quantitative data collected during fitness testing can be compared to national averages.

3.1.3.3 The principles of training and their application to personal exercise/training programmes

Content	Additional information
The principles of training and overload	Key principles of training. SPORT to include: • specificity • progressive overload • reversibility • tedium. Key principles of overload. FITT to include: • frequency • intensity • time • type. Students should be taught the terms and what they mean.
Application of the principles of training	How the principles of training can be applied to bring about improvements in fitness. Application of the principles to sporting examples.

Content	Additional information
Types of training	Understand the distinctions between different types of training. Circuit training – consider space available, equipment available, number of circuit stations, work:rest ratio, the content/demand of the circuit can be altered in order to improve different components of fitness. Continuous training – sustained exercise at a constant rate (steady state) without rests, involving aerobic demand for a minimum of 20 minutes, eg running, swimming, rowing, cycling. Fartlek training – varying speed, terrain and work:recovery ratios. Interval training/high intensity interval training – periods of exercising hard, interspersed with periods of rest or low intensity exercise. Static stretching – a way to stretch to increase flexibility, held (isometric) for up to 30 seconds, using correct technique, advisable to avoid over stretching. Weight training – choice of weight/exercise depends on fitness aim, eg strength/power training or muscular endurance, the importance of safe practice/lifting technique, the need for spotters. Plyometric training – use of plyometric exercises, eg bounding, depth jumping, to increase power. Basic physiological understanding (eccentric contraction followed by larger concentric contraction). Any training (and practice) method must take account of the following: • the training purpose(s), training thresholds/training targets/training zones (see calculating intensities below) • rest/recovery.

Content	Additional information
Identification of the advantages and disadvantages (the effects on the body) of training types linked to specific aims	The advantages and disadvantages (the effects on the body) of each type of training method stated above. Students should be taught to select and evaluate appropriate training methods for various (aerobic and anaerobic) fitness needs and make links to sporting activity, eg continuous training is fully appropriate to marathon runners.

3.1.3.4 How to optimise training and prevent injury

Content	Additional information
Calculating intensities to optimise training effectiveness	Definition of training threshold. Calculate the aerobic/anaerobic training zone: • calculate maximum heart rate (220 minus age) • calculate aerobic training zone (60–80% of maximal heart rate) • calculate anaerobic training zone (80–90% of maximal heart rate). For circuit training, altering the time/rest/content of the circuit will determine the fitness aim. How to calculate one repetition maximum (one rep max) as part of weight training and how to make use of one rep max, with reference to: • strength/power training (high weight/low reps – above 70% of one rep max, approximately three sets of 4–8 reps) • muscular endurance (low weight/high reps – below 70% of one rep max, approximately three sets of 12–15 reps).

Content	Additional information
Considerations to prevent injury	The training type/intensity should match the training purpose (eg aerobic or anaerobic). Where applicable, the following factors should be taken into account in order to prevent injury: • a warm up should be completed • over training should be avoided, eg appropriate weight • appropriate clothing and footwear should be worn • taping/bracing should be used as necessary • hydration should be maintained • stretches should not be overstretched or bounce • technique used should be correct, eg lifting technique • appropriate rest in between sessions to allow for recovery.
Specific training techniques – high altitude training as a form of aerobic training	How high altitude training is carried out: • train at high altitude • there is less oxygen in the air and oxygen carrying capacity is reduced • the body compensates by making more red blood cells to carry oxygen. Students should be taught to evaluate the benefits and the limitations of altitude training for different sports performers. Students do not need to be taught how to calculate intensities for altitude training.

Content	Additional information
Seasonal aspects	Names of the three training seasons: • pre-season/preparation • competition/peak/playing season • post-season/transition. An understanding of what each of the seasons entails (aims): • pre-season/preparation – general/aerobic fitness, specific fitness needs • competition/peak/playing season – maintain fitness levels, work on specific skills • post-season/transition – rest and light aerobic training to maintain a level of general fitness. An understanding of the benefits of each season to the performer. Students should be taught to apply and justify the characteristics of the seasonal aspects to different sporting activities.

3.1.3.5 Effective use of warm up and cool down

Content	Additional information
Warming up and cooling down	The constituent parts of warming up and cooling down. Warming up should include: • gradual pulse-raising activity • stretching • skill based practices/familiarisation • mental preparation • increase amount of oxygen to the working muscles. Cooling down should include: • maintaining elevated breathing and heart rate, eg walk, jog • gradual reduction in intensity • stretching. Students should be taught to understand and justify appropriate elements of a warm up and a cool down for different sporting activities. The benefits of warming up: • effect on body temperature • range of movement increased • gradual increase of effort to full pace • psychological preparation • practice of movement skills through the whole range of movement • injury prevention. The benefits of cooling down: • allowing the body to recover • the removal of lactic acid/CO₂/waste products • prevent (delayed onset) muscle soreness/ DOMS.